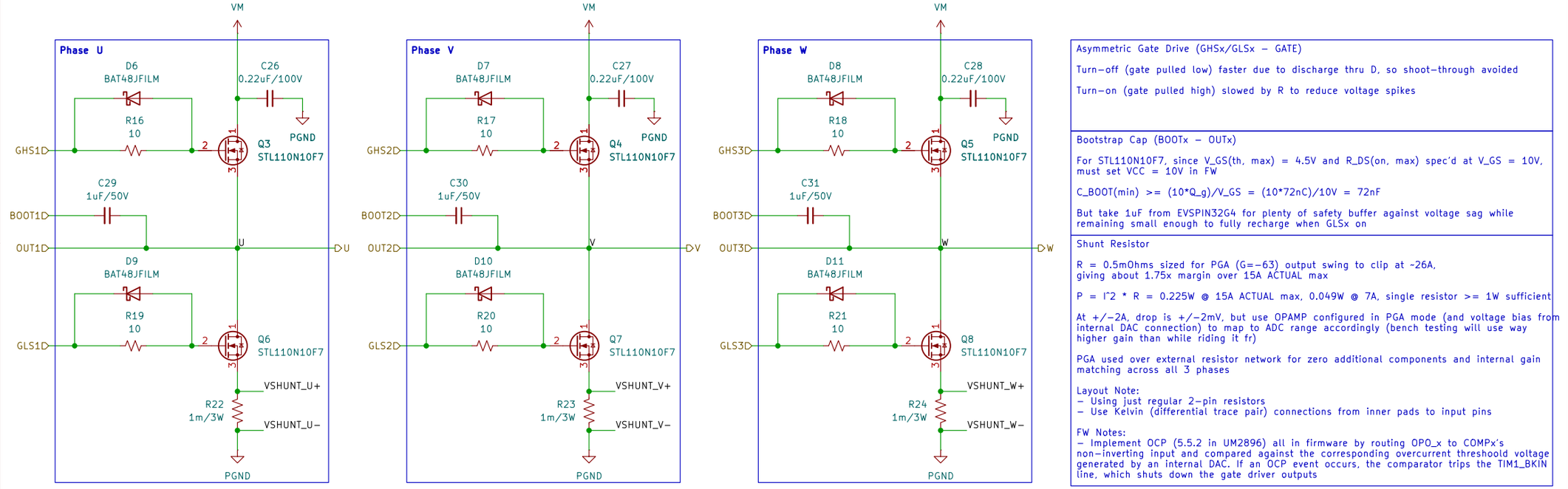


Asymmetric Gate Drive (GHSx/GLSx – GATE)

Turn–off (gate pulled low) faster due to discharge thru D, so shoot–through avoided

Turn–on (gate pulled high) slowed by R to reduce voltage spikes

Bootstrap Cap (BOOTx – OUTx)

For STL110N10F7, since V_GS(th, max) = 4.5V and R_DS(on, max) spec'd at V_GS = 10V, must set VCC = 10V in FW

C_BOOT(min) >= (10*Q_g)/V_GS = (10*72nC)/10V = 72nF

But take 1uF from EVSPIN32G4 for plenty of safety buffer against voltage sag while remaining small enough to fully recharge when GLSx on

Shunt Resistor

R = 0.5mOhms sized for PGA (G=–63) output swing to clip at ~26A, giving about 1.75x margin over 15A ACTUAL max

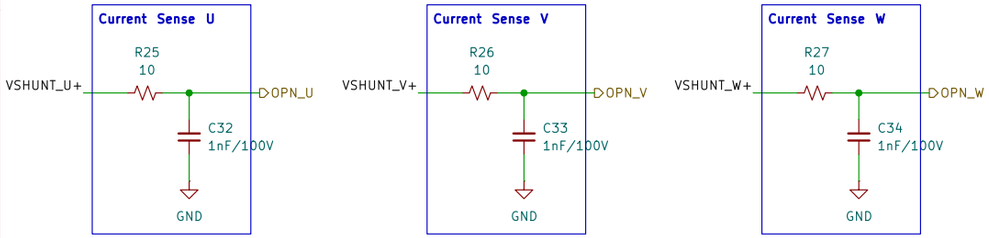
P = I^2 * R = 0.225W @ 15A ACTUAL max, 0.049W @ 7A, single resistor >= 1W sufficient

At +/–2A, drop is +/–2mV, but use OPAMP configured in PGA mode (and voltage bias from internal DAC connection) to map to ADC range accordingly (bench testing will use way higher gain than while riding it fr)

PGA used over external resistor network for zero additional components and internal gain matching across all 3 phases

Layout Note:
– Using just regular 2–pin resistors
– Use Kelvin (differential trace pair) connections from inner pads to input pins

FW Notes:
– Implement OCP (5.5.2 in UM2896) all in firmware by routing OPO_x to COMPx's non–inverting input and compared against the corresponding overcurrent threshoold voltage generated by an internal DAC. If an OCP event occurs, the comparator trips the TIM1_BKIN line, which shuts down the gate driver outputs



PGA Configuration: Inverting Mode (non–inverting disallows DC bias) (AN5306 3.4.1)

VINP (OPP/non–inverting)
– HW: NC, free up OPAMPx_VINP pins. Might need zero–current offset calibration.
– FW: VP_SEL[1:0] = 11 to cut off VINP and route internal DAC ch (RM0440 25.3.4)

VINM (OPN/inverting)
– HW: small R to limit curr (R1_internal=10k => G_error=0.1%)
– FW: VM_SEL[1:0] = 10 to enable PGA mode (internal feedback R's) (AN5306 3.4.1)

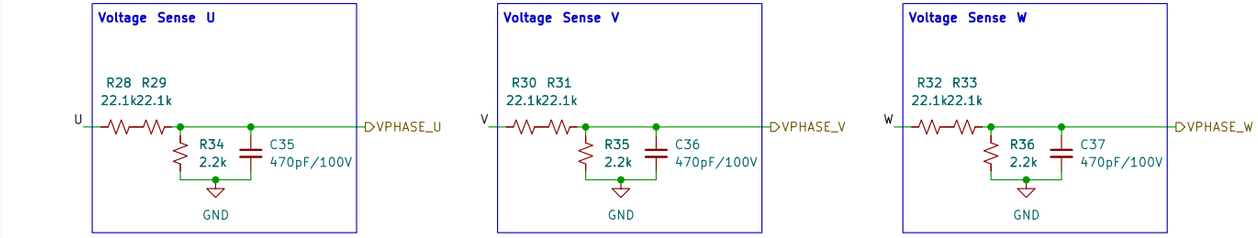
VOUT (OPO)
– HW: NC, free up pins with OPAMPx_VOUT for other periphs
– FW: OPAINTOEN = 1 to cut off VOUT and route internal ADC ch (RM0440 25.3.4)

PGA_GAIN[4:0] = 13 to implement an inverting amplifier (G=–63)

Transfer Function: V_OUT = V_DAC – (|G_inverting| * V_SHUNT)
– To keep 0A midpoint at VDD / 2 = 1.65V, config DAC at 2048
– G=–63: [–26.2, 26.2] [A] –> {0.0, 3.3} [V]

External Filtering Resistor: R_f = 10 ohms
Internal op–amp has high input impedance, so effect on measurement is negligible

External Filtering Capacitor: C_f = 1 nF (C0G/NP0 is a MUST)
STM32G431C6 datasheet gives GBW_min=7MHz & BW_PGA=GBW/G, so G_worst=–63 => BW_worst=7MHz/64=109kHz > f_FOC=50kHz
Theoretically fine to DNP a C_f, but add 1nF anyway to prevent high–freq MOSFET switching noise before captured by the ADC
C_f cannot hurt since f_c=15.91MHz >>> 109kHz=BW_worst



Phase Voltage (=/= BEMF) Sense
– Both FOC and 6–step require the phase voltages to be measured to find BEMF, thus they CAN use the same hardware for measuring phase voltages

6–step Commutation:
– With 6–step, zero–crossings are detected EITHER in–software (by sampling the voltage directly with ADC) OR in–hardware (by using a comparator to detect the crossing)
– This board will only support SOFTWARE–based zero–crossing detection, so the associated pins only require ADCx and not COMPx
– Note: Virtual Neutral Point could also be determined in either SW or HW, but choosing SW (see Control sheet)
– C0G/NP0 is a MUST for the capacitors

R2/(R1+R2) = 0.04741
{60} [V] –> {2.845} [V]

R_eq = R1 || R2 = 2.1k, C_f = 470 pF
=> tau = 0.985 us, f_c = 161.6 kHz

Author: Leopoldo Mendoza, Arnav Shah
Electrium Mobility

Sheet: /Drive/
File: drive.kicad_sch

Title: ESC 2026 – Drive

Size: User	Date: 2026–06–30	**Rev: v2**
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